Note on Bandit Algorithms

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1 Bandits, Probability and Concentration

1.1 Introduction

Definition 1.1.

- *Actions :* $A_1, A_2, \cdots \in \mathcal{A}$ *and we call* $\mathcal{A}$ *the action set.*
- *Rewards :* $X_1, X_2, \cdots \in \mathcal{R}$.
- *History :* $H_{t-1} = (A_1, X_1, \cdots, A_{t-1}, X_{t-1})$.
- *Policy :* $\pi \in \Pi : H_{t-1} \mapsto A_t$ *and we call* Π *the competitor class.*
- *Environment:* $\mathcal{E} : A_t \mapsto X_t$.
- *Regret of a learner relative to a policy* π *until* $n \in \mathbb{N}$ *is*

$$R_n = \left| \sum_{t=1}^{n} X_t - \sum_{t=1}^{n} \mathcal{E}(\pi(H_{t-1})) \right|. \tag{1}$$

where H_0 *represents the initial condition.*

Example 1.2. An environment is called a **stochastic Bernoulli bandit** if

- $\mathcal{A} = \{1, 2, \ldots, k\}$,
- $X_t \in \{0, 1\}$,
- $^{\forall} a \in \mathcal{A}$, $\exists\, \mu_a \in [0,1]^k$ such that $\mathbb{P}[X_t = 1 | A_t = a] = \mu_a$.

Here the optimal policy is to play fixed action $a^* = \operatorname*{argmax}_{a \in \mathcal{A}} \mu_a$. That is, the competitor class Π is the set of all constant policies $\Pi = \{\pi_1, \ldots, \pi_k\}$ where π_i chooses action i in every round. The regret relative to the optimal policy until n rounds is

$$\begin{aligned} R_n &= \left| \sum_{t=1}^{n} X_t - \sum_{t=1}^{n} \mathcal{E}(\pi^*(H_{t-1})) \right| \\ &= \left| \sum_{t=1}^{n} X_t - \sum_{t=1}^{n} \mu_{a^*} \right|. \end{aligned} \tag{2}$$